85/100

CHE 313 Transport Phenomena III

Exam #2 (10/30/2025)

Name: Diata

1. (100 pts) The following saturated liquid feed mixture (Table) is to be separated by a distillation column at 120 psia to obtain 95.0 mol% of pentane (C_5) in the distillate and 96.0 mol% of the hexane (C_6) in the bottom. The column has a partial condenser and a partial reboiler.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

Component	$\alpha_{i\text{-ref}}$	kmol/hr
C3, propane	4.36	5
C4, butane	2.36	15
C5, pentane	1.88	25
C6, hexane	1.00	35
C7, heptane	0.84	20
Total in feed	-	100

1) (10 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
2) (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK.
3) (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
4) (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for mole fraction using stoichiometric condition.
5) (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln\left\{\frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,Bot}}{FR_{B,dist}}\right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln\left\{\frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A}\right\}}{\ln \alpha_{AB}}$$

where A=LK and B=HK, $FR_{i,dist} = (Dx_{i,dist})/Fz_i$ & $FR_{i,Bott} = (Bx_{i,Bot})/Fz_i$

$q = 1$

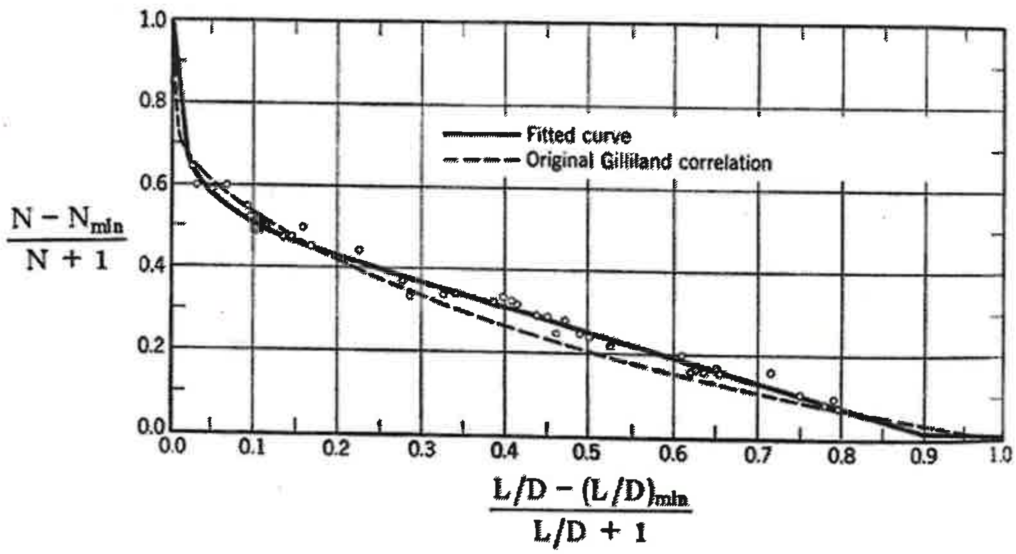
6) (15 pts - Underwood equation) After drawing mass balance envelope around the feed stage, find ΔV_{feed}. Then, use the following equation to set up equation for ϕ value calculation, so you don't need to calculate it. Define the required all variables and expand the equation from i = 3 – 7. Here, ref = HK. Four ϕ values (0.8843, 1.354, 2.175, and 4.001) are obtained. Which one is correct?

$$\Delta V_{feed} = \sum_{i=1}^{c} \frac{\alpha_{i-ref} F z_i}{\alpha_{i-ref} - \varphi}$$

7) (10 pts- Underwood equation) Find V_{min} and $(L/D)_{min}$. Use mass balance between L_{min}, V_{min}, and D.

$$V_{min} = \sum_{i=1}^{c} \frac{\alpha_{i-ref}(Dx_{i,dist})}{\alpha_{i-ref} - \phi}$$

8) (10 pts-Gilliland correlation) If R/R_{min} = 1.3, find N. Use the following the Fitted correlation curve to calculate N. Use N_{min} and $N_{F,min}$ from (5).



9) (5 pts) Find N_F by using the following ratio equation.

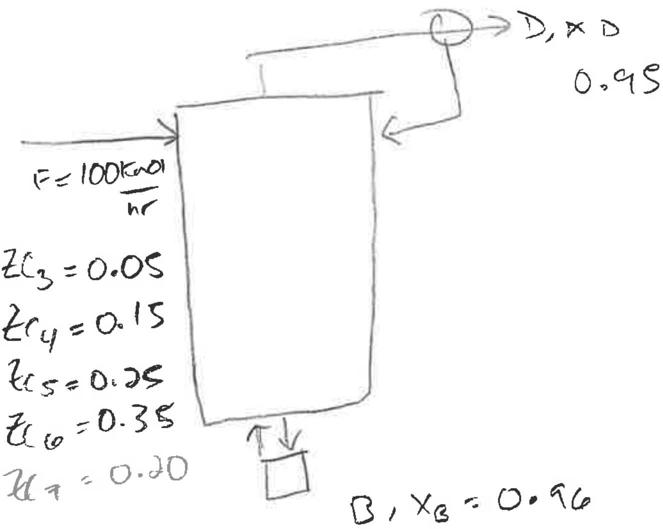
$$\frac{N_{F,min}}{N_{min}} = \frac{N_F}{N}$$

10) (5 pts) Find N_F by using Kirkbride's equation. Compare your result to the value from (9).

$$\left(\frac{N_F - 1}{N - N_F}\right) = \left[\frac{B}{D}\left(\frac{z_{HK}}{z_{LK}}\right)\left[\frac{x_{LK,bot}}{x_{HK,Dist}}\right]^2\right]^{0.206}$$

11) (5 pts) Explain whether this distillation process is reasonably economical.

1) $x_{C5/D} = 0.95$
$x_{C6/B} = 0.96$ these ARE $f_R x_i$!! Pg1



$\alpha_{C3-C6} = 4.36$
$\alpha_{C4-C6} = 2.36$
$\alpha_{C5-C6} = 1.88$
$\alpha_{C6-C6} = 1.00$
$\alpha_{C7-C6} = 0.84$

1) 10/10

$F = B + D$
$Fz_i = Bx_{i,B} + Dx_{i,D}$

$f_{C5} = 0.95$ $f_{C6} = 0.96$

2) 5/5

2) LNK: C_4 & C_3 | LK: C_5 | NK | HK: C_6 | HNK: C_7

$F = B + D$ $100 = B + D$

3) Distillate
$Dx_i = F \cdot z_i \cdot f_R$
$Dx_{i,D}$

$Dx_{C7} = 0$

C_6: $Dx_{C6,D} = 100(0.35)(1-0.96) = 1.4 \frac{kmol}{hr}$

C_5: $Dx_{C5} = 25 \frac{kmol}{hr}(0.95) = 23.75 \frac{kmol}{hr}$

3) 15/15

C_4: $Dx_{C4} = 100(0.15) = 15 \frac{kmol}{hr}$

C_3: $Dx_{C3} = 100(0.05) = 5 \frac{kmol}{hr}$

$D = 45.155 \frac{kmol}{hr}$

Mole fractions

$x_{C5,D} = 0.526$ $x_{C4,D} = 0.33$ $x_{C3,D} = 0.1107$

$x_{C6,D} = 0.031$ $x_{C7,D} = 0$

Distillate

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$C_7: Bx_{B,C_7} = (F_{C7})(z_{C7}) \Rightarrow Bx_{B,C_7} = 20 \frac{kmol}{hr}$

$C_5: Dx_{B C_5} = 25 \frac{kmol}{hr}$

$C_6: B(0.96) = 100(0.35) =$

$B = \frac{35 \frac{kmol}{hr}}{0.96} = 36.45$

$x_{B,C_5} = 0.96$

$Dx_{C_6} = F \cdot z \cdot f_R$

$C_6: Dx_{C_6,D} =$

3) Bottom

$C7: Bx_{C7,B} = 100(0.20) = 20 \frac{kmol}{hr}$

$C6: Bx_{C7,B} = 100(0.35)(0.96) = 33.6 \frac{kmol}{hr}$

$C_5: Bx_{C5,B} = 25 \frac{kmol}{hr}(1-0.95) = 1.25 \frac{kmol}{hr}$

$B = 54.85 \frac{kmol}{hr}$

$x_{C7,B} = 0.364$

$x_{C5,B} = 0.0227$

$x_{C6,B} = 0.612$

4) 15/15

Cs:
$25 \frac{kmol}{hr} =$

$\frac{1}{2} 25 \text{ kmol/hr} = B x_{i,B} + D$

$F = B + D$
$D = F - B$
$B = L = \left(\frac{L}{D}\right) D$

PG 5

7) 5/10

7)

$$V_{min} = \frac{4.36(5')}{4.36 - 0.8843} + \frac{2.36(15)}{2.36 - 0.8843} + \frac{1.88(23.75)}{1.88 - 0.8843} + \frac{1(1.4)}{1 - 0.8843} + 0$$

(Cs) (C4) (C5) NO C7 in Dist

$V_{min} = 6.27 + 23.98 + 44.84 + 12.10$

$V_{min} = \cancel{87.19}$ 123.37 $\frac{kmol}{hr}$

$D = 45.155$

$L_{min} = V_{min} - D$

$L_{min} = \cancel{42.035}$ 78.22 kmol/hr

unit missing

$\frac{L}{D_{min}} = \cancel{0.93}$ 1.73

8)

$$\frac{\frac{L}{D}}{\frac{L}{D_{min}}} = 1.3$$

$X = \frac{R - R_{min}}{R+1}$

$0.93(1.3) = \frac{L}{D}$

$\frac{L}{D} = 1.209$

$Y = \frac{N - N_{min}}{N+1}$

9/8

$$X = \frac{1.209 - 0.93}{1.209 + 1}$$

$N =$

$X = 0.126$ From graph $Y \approx 0.5$

(6)

6) M.B envelope ×

Pg 4

Explain!

$\Delta V_{feed} = F(1-q) =$

$$0 = \frac{\alpha_{C3-C6}\cancel{F} z_{C3}}{\alpha_{C3-C6} - \phi} + \frac{\alpha_{C4-C6}\cancel{F} z_{C4}}{\alpha_{C4-C6} - \phi} + \frac{\alpha_{C5-C6}\cancel{F} z_{C5}}{\alpha_{C5-C6} - \phi}$$

6) 5/15

$$+ \frac{\alpha_{C6-C6}\cancel{F} z_{C6}}{\alpha_{C5-C6} - \phi} + \frac{\alpha_{C7-C6}\cancel{F} z_{C6}}{\alpha_{C7-C6} - \phi}$$

$$= \frac{4.36(0.09)}{4.36-\phi} + \frac{0.36(.015)}{0.36-\phi} + \frac{1.88(.25)}{1.88-\phi} + \frac{1(.35)}{1-\phi} + \frac{0.84(.20)}{0.84-\phi}$$

$\phi = 1.354$

since $\alpha_{HK} < \phi < \alpha_{LK}$

ϕ is where equation $= 0$

$\phi = 4.001$

closest to 0

$\phi = \cancel{2.175}$ 0.8843

this is not Phi

$$\frac{4.36(0.05)}{1-4.001}$$

-0.0072

math error

$$\frac{4.36(0.05)}{1-1.354} = 0.6158$$

$$\frac{4.36(0.05)}{(1-1.354)} = -0.18$$

$$\frac{4.36(0.05}{1-0.8843} = 1.88$$

5) $N_{F\,min} = \dfrac{\ln\left[\dfrac{X_{C5,D}}{X_{C6,D}} \times \dfrac{z_{C6}}{z_{C5}}\right]}{\ln \alpha_{C5-C6}}$

PG 3

5) 5/5 $N_{F\,min} = \dfrac{\ln\left[\dfrac{0.526}{0.031} \times \dfrac{0.35}{0.25}\right]}{\ln(1.88)} = \dfrac{\ln\left[16.96 \times 1.4\right]}{\ln(1.88)} = \dfrac{3.167}{0.631}$

$N_{F\,min} = 5$ 5 stages

$N_{min} = \dfrac{\ln\left[\dfrac{0.95}{1-0.95} \times \dfrac{0.96}{(1-0.96)}\right]}{\ln(1.88)}$

$= \ln\left(\dfrac{0.95}{0.05} \times \dfrac{0.96}{0.04}\right)$

$\dfrac{\ln[456]}{\ln(1.88)} = 4.698$

10 stages

9 cont

$$Y = \frac{N - N_{min}}{N+1} \Rightarrow 0.5 = \frac{N-10}{N+1}$$

P96

$$0.5(N+1) = N - 10$$

$$0.5N + 0.5 = N - 10$$

$$0.5N = 10.5$$

$$N = \cancel{21} \; 20 \quad \text{Total \# stages}$$

Method B OK!

8) 9/10

9) 10/10

10) 4/5

11) 2/5

10) $B = 54.85 \quad D = 115.155$

$$= \left(\left[\frac{54.85}{115.155} \left(\frac{0.35}{0.25} \frac{z_{C6}}{z_{C5}} \right) \left(\frac{0.0227}{0.031} \right) \right]^2 \right]^{0.206}$$

B, D; $x_{C5,bot}$; $x_{C6,D}$

$$= \left[\left[1.21 \, (1.4)(0.732) \right]^2 \right]^{0.206}$$

$= \cancel{1.09}$ 10.01

-1

vs

9) $N_F = N \dfrac{N_{F\,min}}{N_{min}} = 10.5$

$$\frac{N_F - 1}{N - N_F} = 0.404$$

11) Not economical ~~they~~ aren't the same total amount of stages but are close